CMPG 325 – COMPUTER NETWORKS

MILESTONE 3

PROJECT : Network Design for Botsho Geotechnical Engineers

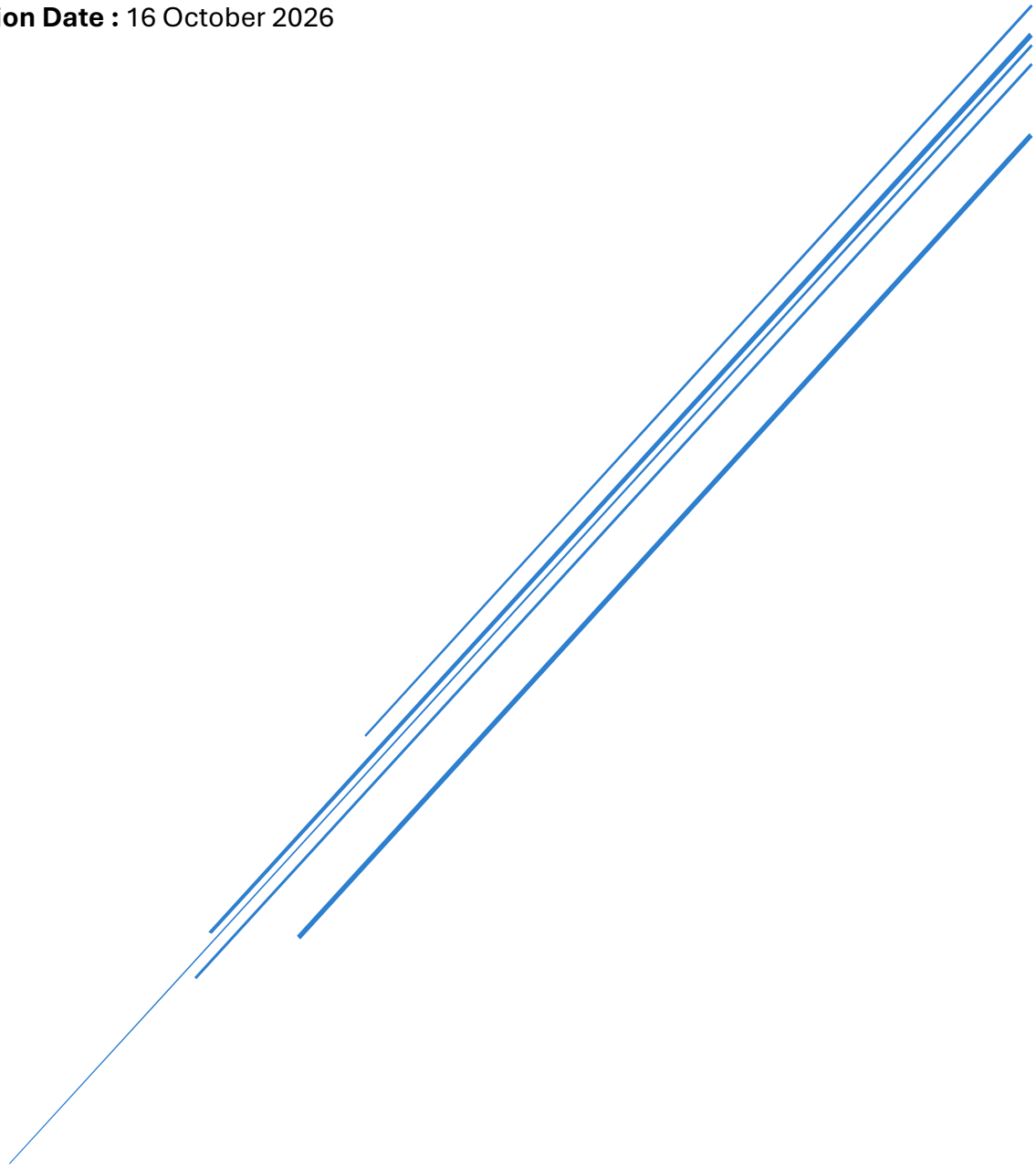
PROJECT ID: CMPG325-2026-068

CLIENT ID: CLI-068

ORGANISATION : Botsho Geotechnical Engineers (Potchefsroom)

Industry: Engineering

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Executive Summary

This technical report presents the completed design, implementation, verification and documentation of a Cisco Packet Tracer network for Botsho Geotechnical Engineers. The project uses the assigned 172.30.42.0/23 IPv4 block and addresses the assigned technical challenge of DHCP with scoped multi-VLAN address assignment. It also accommodates Change Request CR8, which requires a shared printer zone to serve two departments, and the design constraint that critical file, print and application services remain accessible during business hours.

The implemented solution uses four VLANs: Engineering (VLAN 10), Administration (VLAN 20), Shared Printer (VLAN 30) and Servers (VLAN 40). A router-on-a-stick architecture provides inter-VLAN routing over an IEEE 802.1Q trunk. Engineering and Administration clients receive IPv4 settings dynamically from separate DHCP scopes, while the printer and server use static addresses for predictable access. The solution was validated through VLAN and trunk verification, DHCP pool and binding checks, client IP configuration checks, and end-to-end ping tests.

The final evidence demonstrates multi-VLAN DHCP operation, inter-VLAN communication, printer reachability from both required departments and server reachability. The design is appropriate to the scope of the simulated client network. It supports predictable access to critical services.

Report Structure

- Client requirements and project scope
- Physical and logical network design
- IPv4 addressing and VLSM plan
- Packet Tracer implementation and configuration
- Assigned DHCP technical challenge
- CR8 shared-printer solution

1. Introduction and Project Context

1.1 Client Background

The assigned client is Botsho Geotechnical Engineers, an engineering organisation located in Potchefstroom. The project required the analysis, design and simulation of a computer network in Cisco Packet Tracer. Engineering and Administration were used as the two departmental user groups for the purpose of demonstrating departmental segmentation, scoped DHCP and the shared-printer requirement.

1.2 Project Objectives

The project objectives were to design a topology, use the assigned addressing block as the basis for the IPv4 plan, configure the required routers, switch and end devices, implement the assigned networking challenge, satisfy the client change request, demonstrate end-to-end connectivity, and maintain a portfolio of evidence showing the development and testing process.

1.3 Project-Specific Requirements

Requirement	Assigned detail
Addressing block	172.30.42.0/23; subdivided using VLSM
Technical challenge	DHCP with scoped multi-VLAN address assignment
Change request	CR8: a shared printer zone must serve two departments
Design constraint	Critical file/print/application services must remain available during business hours
Simulation platform	Cisco Packet Tracer
Evidence platform	Individual GitHub portfolio plus technical documentation and screenshots

2. Client Requirements Analysis

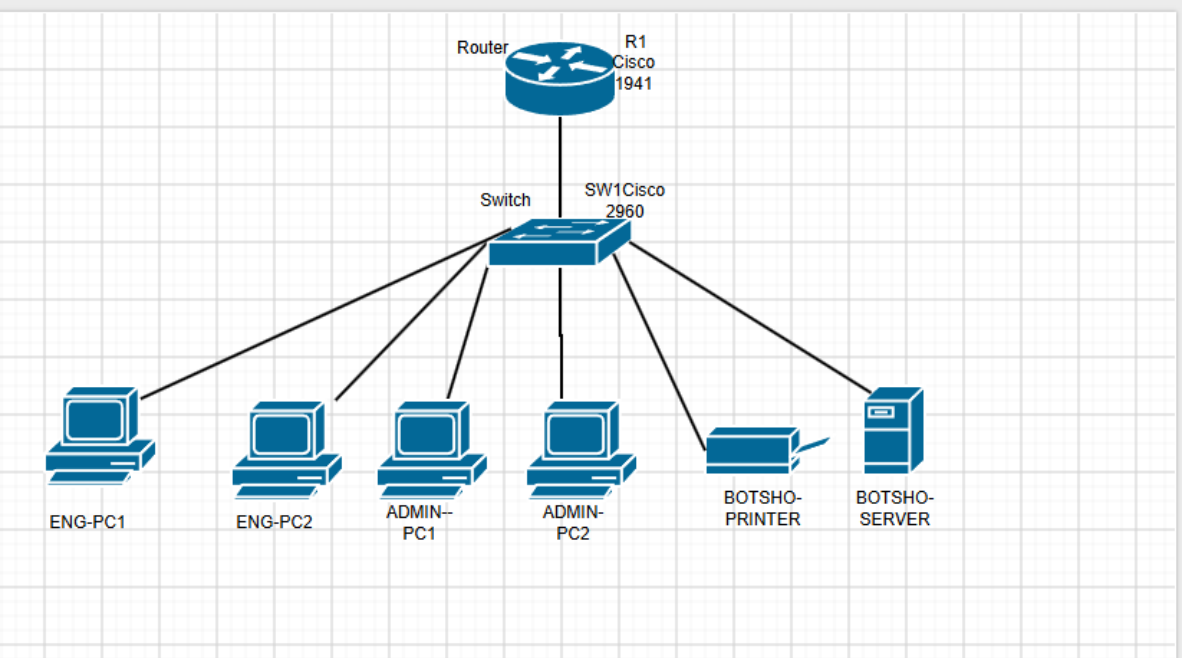
The solution was designed around the assigned requirements. The network must provide appropriate connectivity between departmental users and shared resources, maintain logical separation between network functions, allocate client addressing correctly, and allow the two departmental networks to access the shared printer. The final implementation preserves these requirements in Packet Tracer.

Client need	Network response
Departmental connectivity	Engineering and Administration clients connected through SW1.
Logical separation	Four VLANs separate departmental users from shared resources.
Inter-VLAN communication	R1 provides router-on-a-stick routing between VLANs.
Automatic departmental addressing	Separate DHCP scopes serve Engineering and Administration.
Shared printing	BOTSHO-PRINTER is placed in VLAN 30 and reached through inter-VLAN routing.
Predictable service addresses	Printer and server use static IPv4 addresses.
Critical service access	Dedicated resource VLANs and routed connectivity provide predictable access during normal operation.
Evidence and reproducibility	Configurations, screenshots, tests and reports are maintained in the project portfolio.

3. Network Design

3.1 Physical Topology

The physical design uses a centralised star arrangement. SW1 acts as the primary Layer 2 connection point for end devices, while R1 connects to the switch and provides the routed gateway and DHCP functions. In the final Milestone 2 implementation, R1 is a Cisco 2911 router. Four departmental client PCs, one network printer and one server are used to demonstrate the required functions.



During Milestone 1, I used a Cisco 1941 router as part of the proposed network design. When I moved to the actual Packet Tracer implementation in Milestone 2, I selected a Cisco 2911 router as the final device. The 2911 supported the functions required for my implementation, including Gigabit Ethernet connectivity, 802.1Q subinterfaces for router-on-a-stick, inter-VLAN routing and DHCP. The change was only a hardware-model change; it did not affect the logical design, VLAN structure, IP addressing plan or DHCP scopes.

Device	Quantity	Purpose
R1 - Cisco 2911 Router	1	Inter-VLAN routing and DHCP services; final Milestone 2 router model
SW1 - Cisco 2960 Switch	1	VLAN switching, access ports and 802.1Q trunking
Engineering PCs	2	DHCP clients in VLAN 10
Administration PCs	2	DHCP clients in VLAN 20

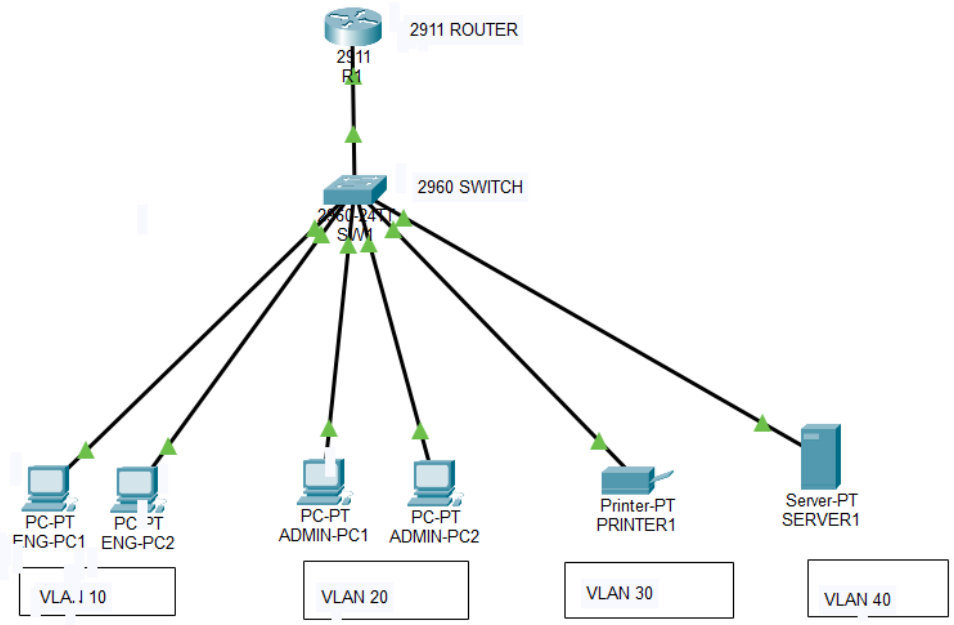
BOTSHO-PRINTER	1	Shared printer resource in VLAN 30
BOTSHO-SERVER	1	Static server resource in VLAN 40

3.1.1 Router Model Change Between Milestones

Implementation update: The Milestone 1 design diagram initially represented R1 as a Cisco 1941 router. During Milestone 2, the network was implemented using a Cisco 2911 router instead. This was a device-model change only and did not alter the approved logical design, VLAN structure, VLSM addressing plan, router-on-a-stick approach, DHCP scopes or testing objectives. The Cisco 2911 therefore became the router used in the final Packet Tracer implementation.

3.2 Logical Topology

The logical design divides the network into four broadcast domains. VLAN 10 contains Engineering users, VLAN 20 contains Administration users, VLAN 30 provides the shared-printer zone required by CR8, and VLAN 40 contains server resources. R1 provides the Layer 3 path between these VLANs while preserving their Layer 2 separation.



3.3 Switch Port Allocation

SW1 Port	Connected Device	Mode	Access
Gi0/1	R1	802.1Q trunk	10, 20, 30, 40
Fa0/1	ENG-PC1	Access	10
Fa0/2	ENG-PC2	Access	10

Fa0/3	ADMIN-PC1	Access	20
Fa0/4	ADMIN-PC2	Access	20
Fa0/5	BOTSHO-PRINTER	Access	30
Fa0/6	BOTSHO-SERVER	Access	40

4. IP Addressing and VLSM Design

4.1 Assigned Address Block

The assigned client block is 172.30.42.0/23 with subnet mask 255.255.254.0. The /23 covers 172.30.42.0 through 172.30.43.255. Because the client design requires several independent VLANs, the /23 was subdivided using Variable Length Subnet Masking (VLSM).

4.2 VLSM Allocation

VLAN	Purpose	Network	Mask	Usable range	Broadcast
10	Engineering	172.30.42.0	/26 | 255.255.255.192	172.30.42.1-62	172.30.42.63
20	Administration	172.30.42.64	/26 | 255.255.255.192	172.30.42.65-126	172.30.42.127
30	Shared Printer	172.30.42.128	/28 | 255.255.255.240	172.30.42.129-142	172.30.42.143
40	Servers	172.30.42.144	/28 | 255.255.255.240	172.30.42.145-158	172.30.42.159

The /26 departmental subnets provide 62 usable host addresses each, allowing substantially more users than the four PCs used in the demonstration. The /28 printer and server subnets provide 14 usable addresses each, which is sufficient for smaller service zones while conserving the assigned address space. The remainder of the /23 stays available for future expansion.

5. Packet Tracer Implementation

5.1 VLAN and Trunk Configuration

VLANs 10, 20, 30 and 40 were created on SW1. The corresponding access ports were assigned to the appropriate departmental or service VLAN. The uplink from SW1 to R1 was configured as an IEEE 802.1Q trunk and the required VLANs were allowed across the trunk. This allows a single physical link to transport frames belonging to all four VLANs while maintaining logical separation.



SW1 verification showing the 802.1Q trunk and VLAN 10/20/30/40 port assignments.

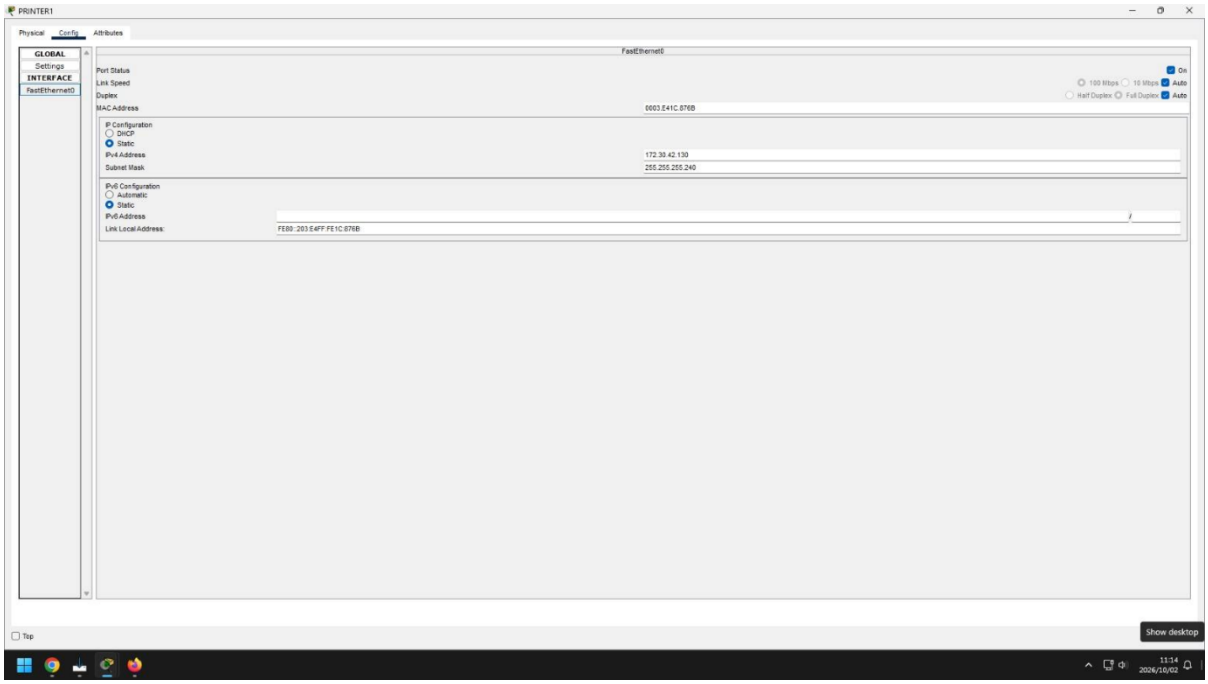
5.2 Router-on-a-Stick Inter-VLAN Routing

In the final Milestone 2 implementation, R1 is a Cisco 2911 router and provides a default gateway for each VLAN through router subinterfaces. The verified gateway addresses are 172.30.42.1, 172.30.42.65, 172.30.42.129 and 172.30.42.145. Each subinterface is associated with its VLAN using 802.1Q tagging. Traffic destined for another VLAN is sent by the host to its local gateway, routed by R1, and returned over the trunk with the appropriate destination VLAN tag.

Router-on-a-stick was selected because it provides a clear and efficient way to demonstrate inter-VLAN routing in a small simulated network that uses one router and one Layer 2 switch.

5.3 Static Resource Addressing

The printer and server were assigned static IPv4 addresses so that shared services remain at predictable network locations. BOTSHO-PRINTER uses 172.30.42.130/28 with gateway 172.30.42.129, while BOTSHO-SERVER uses 172.30.42.146/28 with gateway 172.30.42.145. Static service addresses simplify user access, documentation and troubleshooting.



Shared printer static IPv4 configuration: 172.30.42.130/28.

6. Assigned Networking Challenge: Scoped Multi-VLAN DHCP

6.1 Technical Requirement

The assigned technical challenge requires DHCP to be configured, verified and demonstrated across multiple VLANs. The objective is not merely to allocate addresses automatically; each departmental client must receive an address, subnet mask and default gateway that belong to the correct logical network.

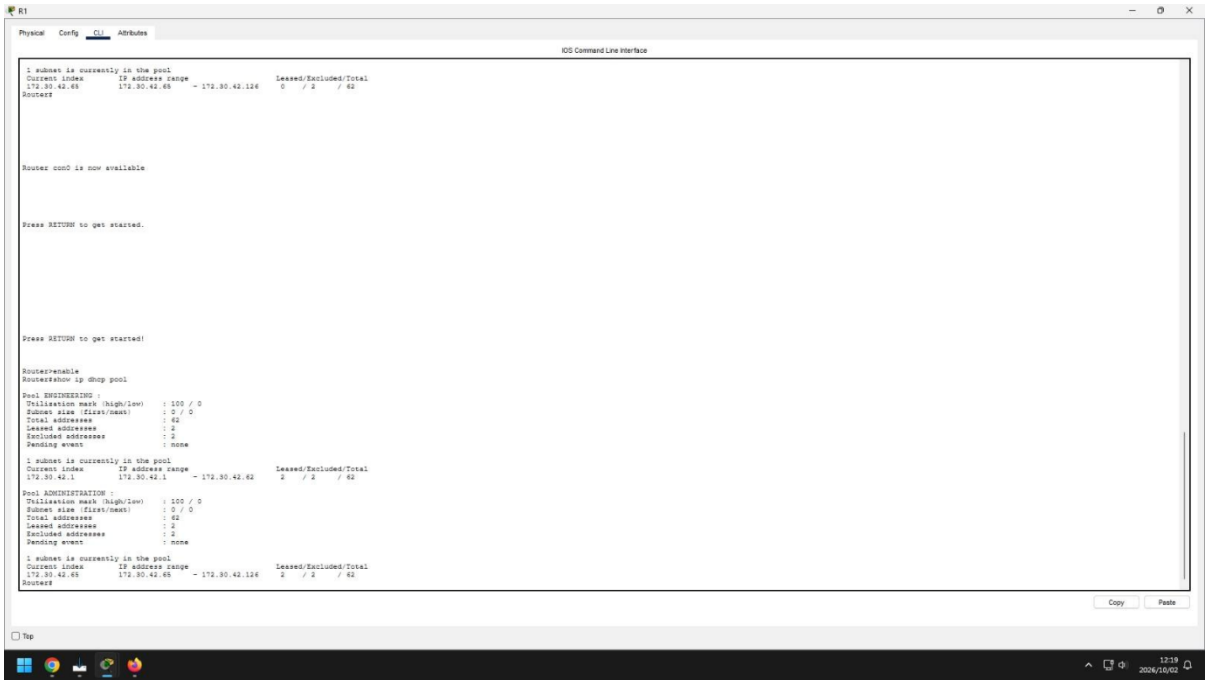
6.2 DHCP Scope Design

DHCP Pool	VLAN	Network	Default Gateway	Dynamic Client Range
ENGINEERING	10	172.30.42.0/26	172.30.42.1	172.30.42.10-172.30.42.62
ADMINISTRATION	20	172.30.42.64/26	172.30.42.65	172.30.42.70-172.30.42.126

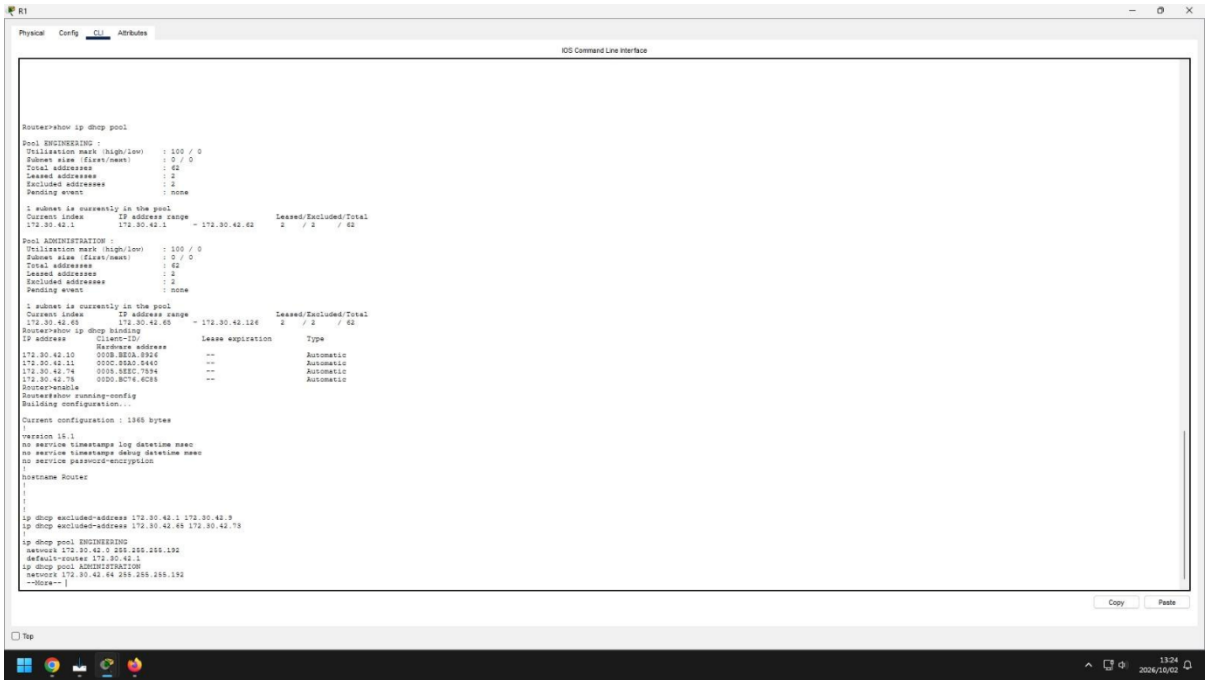
Gateway and infrastructure addresses were protected through DHCP exclusions. Engineering reserves 172.30.42.1 through 172.30.42.9, and Administration reserves 172.30.42.65 through 172.30.42.69. This prevents dynamically assigned client addresses from conflicting with fixed gateway or infrastructure addresses.

6.3 DHCP Verification

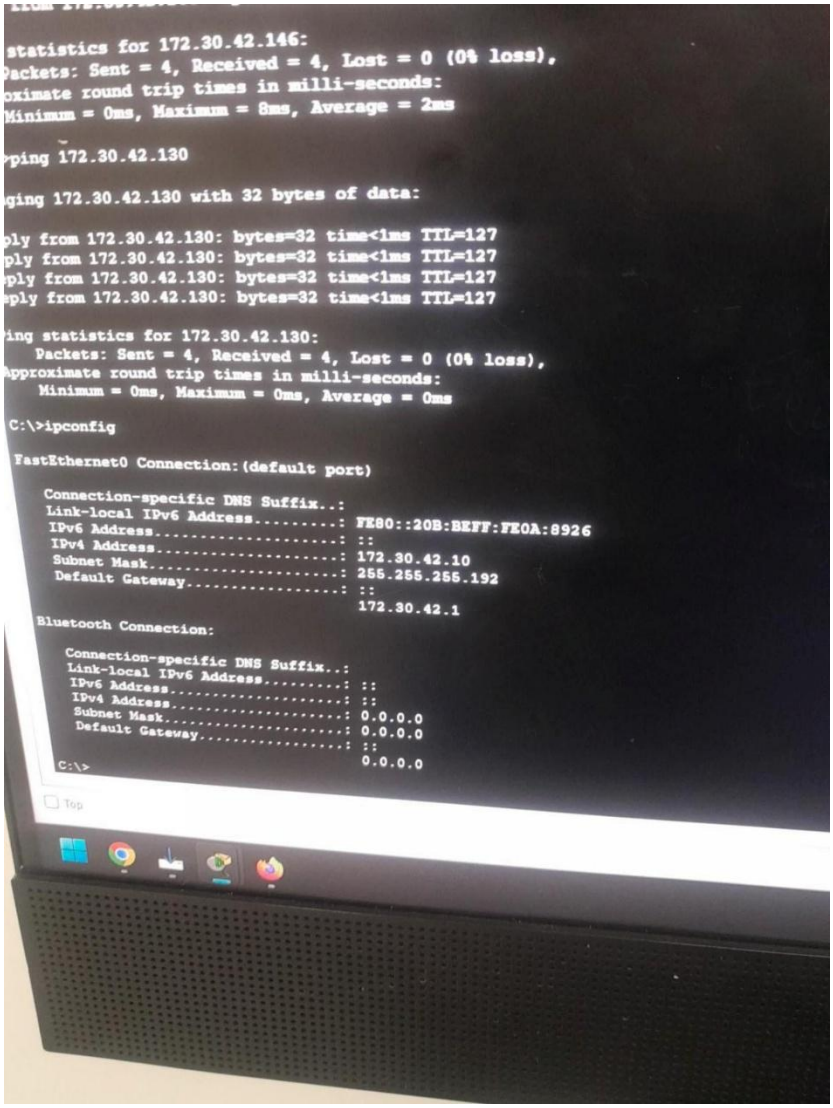
The DHCP implementation was verified from both the router and client perspectives. Router pool output confirms that the two scopes exist, and the DHCP binding table confirms that leases were issued. Observed leases include 172.30.42.10 and 172.30.42.11 in Engineering, and 172.30.42.74 and 172.30.42.75 in Administration. Client-side IP configuration confirms that the received masks and gateways match the intended departmental subnet.



Router DHCP pool status for Engineering and Administration.



DHCP pool, binding and running-configuration evidence.



Engineering client IPv4 configuration showing DHCP address 172.30.42.10/26 and gateway 172.30.42.1.

6.4 The DHCP Evidence

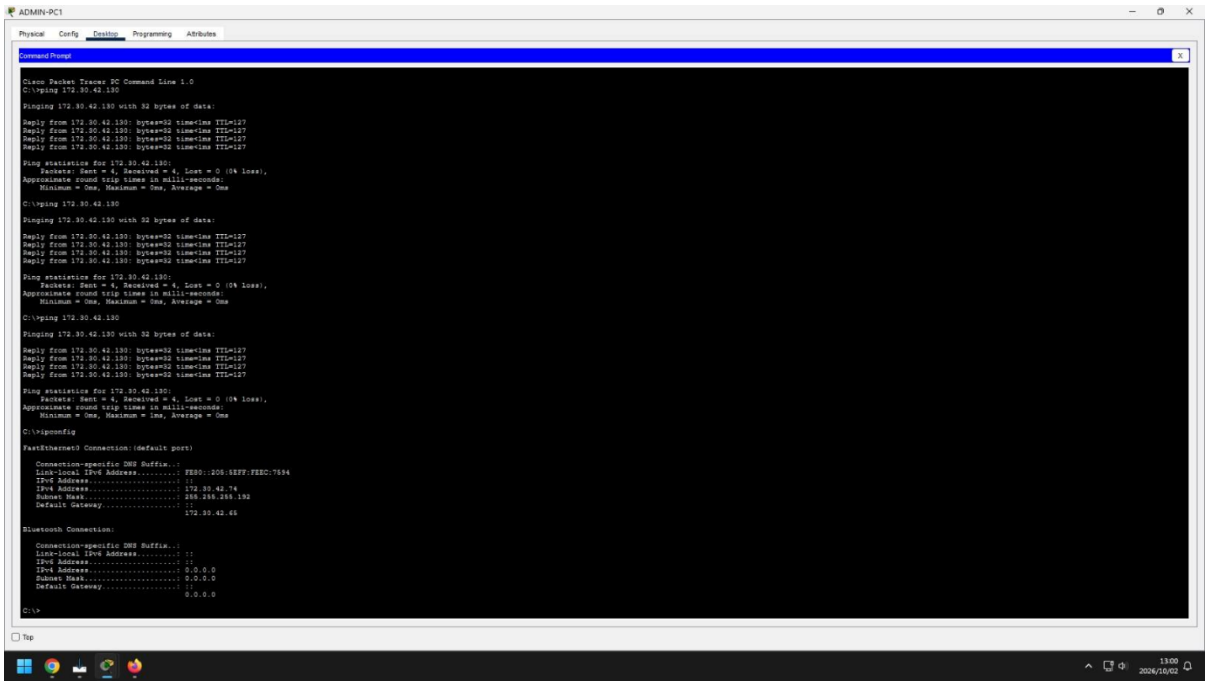
The combination of pool status, binding information and client-side configuration verifies the complete process. The router shows that addresses were allocated from the configured scopes, while the clients show that the correct IPv4 information was actually received. This demonstrates scoped multi-VLAN address assignment rather than only showing that DHCP commands were entered.

7. Client Change Request CR8: Shared Printer Zone

CR8 requires a shared printer zone to serve two departments that currently cannot print. The final design satisfies this requirement by placing BOTSHO-PRINTER in a dedicated VLAN 30 rather than assigning it directly to Engineering or Administration. This treats the printer as a shared organisational resource while preserving the logical separation of the two user VLANs.

The printer uses the static address 172.30.42.130/28 and default gateway 172.30.42.129. Inter-VLAN routing through R1 provides the path from VLAN 10 and VLAN 20 to the printer subnet.

Successful ping tests from both departments demonstrate that the change request is functional, not merely configured.



Administration-to-printer reachability test showing successful replies and 0% packet loss.

8. Critical Services Design Constraint

The project specifies that critical file, print and application services must remain available during business hours. Within the scope of the Packet Tracer simulation, predictable access is supported by placing the printer and server in dedicated service VLANs, assigning them static IPv4 addresses, and providing routed access from the departmental VLANs.

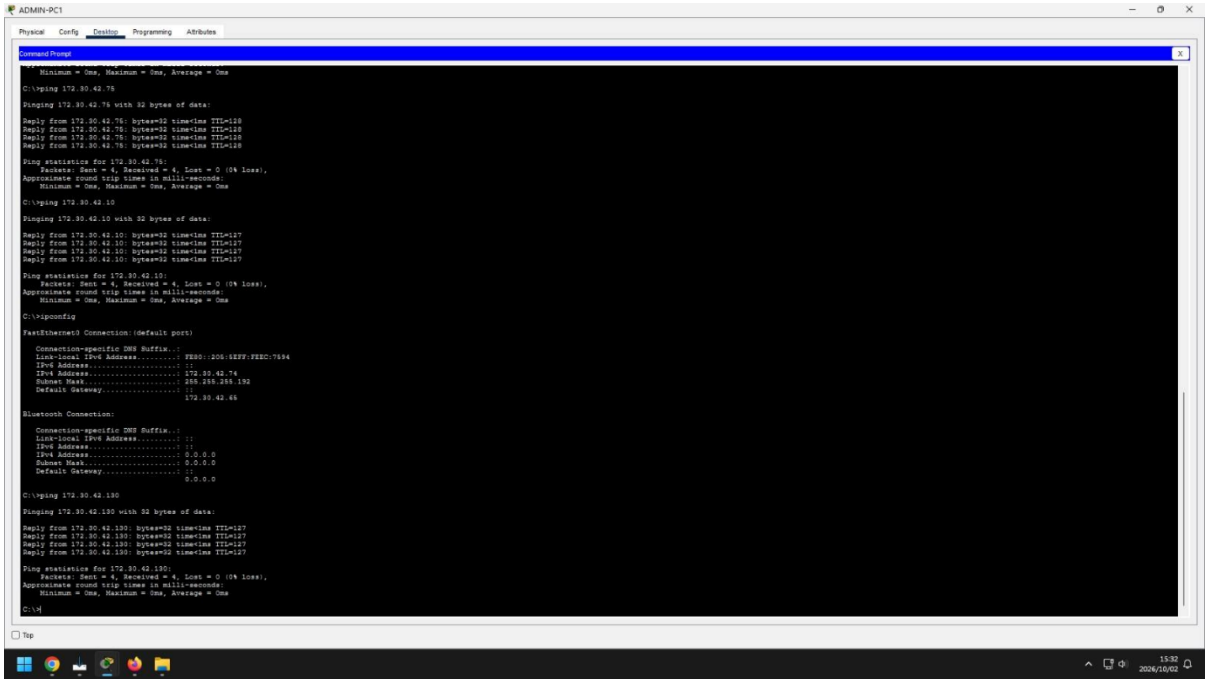
9. Testing and Verification

9.1 Testing Method

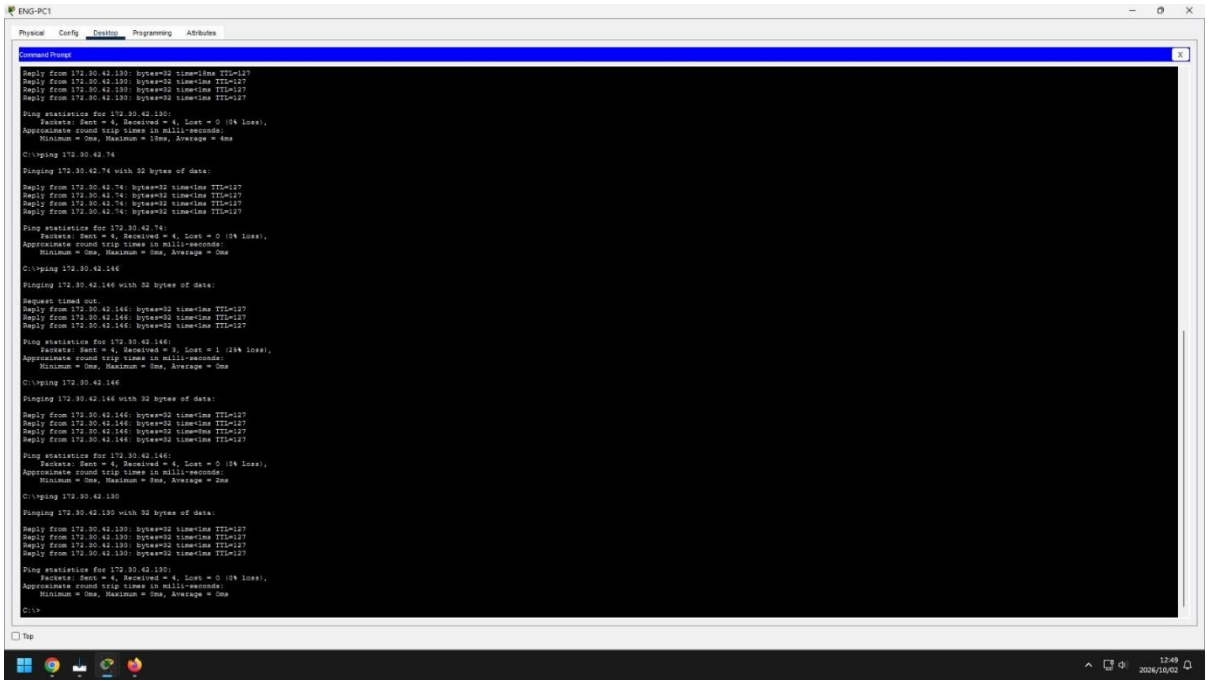
Testing was performed progressively so that faults could be isolated by layer. The process began by checking host addressing and local gateway reachability, followed by inter-VLAN communication, shared-printer access and server reachability. Switch VLAN membership, trunk status, router interfaces, DHCP pools and DHCP bindings were also verified using device commands.

Test	Expected Results	Observed results	Conclusion
SW1 VLAN membership	Ports assigned to VLANs 10, 20, 30 and 40	Verified	VLAN segmentation implemented
SW1 trunk	Gi0/1 trunking; VLANs 10,20,30,40 allowed	Verified	Trunk operational

Engineering DHCP	Client receives 172.30.42.0/26 address and gateway .1	Verified	Engineering scope operational
Administration DHCP	Client receives 172.30.42.64/26 address and gateway .65	Verified	Administration scope operational
Inter-VLAN communication	Engineering and Administration hosts can communicate	Successful	Routing operational
Engineering -> printer	Reach 172.30.42.130	Successful	CR8 path operational
Administration -> printer	Reach 172.30.42.130	Successful	CR8 path operational
Client -> server	Reach 172.30.42.146	Successful after repeat verification	Server VLAN reachable
DHCP binding check	Leases displayed for both client VLANs	Verified	Scoped DHCP demonstrated



Administration client verification showing DHCP configuration, inter-VLAN connectivity and shared-printer reachability.



Engineering client connectivity tests to Administration, the server and the shared printer.

9.2 Interpretation of Results

The final tests confirm that the addressing, VLAN membership, trunking and routing components work together. A successful DHCP lease alone would not prove that the client could reach other networks, while a successful ping alone would not prove that DHCP was scoped correctly. The combined evidence therefore validates both configuration and operation.

During server testing, one initial ping attempt contained a timeout, while the repeated test immediately afterwards completed with 0% packet loss. The final verified path was therefore operational. Repeating the test after the network path had resolved was included as part of the validation process rather than relying on the first result.

10.1 Scalability

The design uses only a portion of the assigned /23 block. The user subnets contain considerably more capacity than the four client PCs used in the demonstration, and additional portions of the assigned address space remain unused. This allows future devices or additional VLANs to be introduced without redesigning the existing address plan.

10.1. Conclusion

The Botsho Geotechnical Engineers network project progressed from client requirements and network design to Packet Tracer implementation and verification. The assigned 172.30.42.0/23 block was subnetted efficiently using VLSM and divided into four VLANs for Engineering, Administration, the shared printer and server resources.

Router-on-a-stick provides inter-VLAN communication through an 802.1Q trunk. Separate DHCP pools successfully provide scoped address assignment to Engineering and

Administration clients, while static addressing gives the printer and server predictable locations. CR8 is satisfied because both departmental networks can reach the shared printer in VLAN 30. Testing also demonstrates inter-VLAN communication and server reachability.

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